

Cowboys

FACEROLL

TEST PLAN

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Version 1.1

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Introduction

This Test Plan will be used to define the strategy for testing FaceRoll. This document will communicate the overall test objectives, scopes, risks, and deliverables for the team. It will also identify what will be tested during each sprint and how the team will verify that the system meets its functional and non-functional requirements. The overall goal of testing is to ensure the project performs reliably as a classroom tool.

Objectives

The objective of this test plan is to verify that all major system features are operating correctly across the student mobile app, instructor dashboard, Firebase backend, and local recognition system.

Testing shall focus on confirming all users will be able to work through authentication such as register and log in, students can enroll their faces, and instructors can start and manage attendance sessions, the local recognition system should be able to identify students, and send results back to the dashboard so they can be stored within the database, also instructor can review and manually update their records when needed, The goal is make FaceRoll and reliable, usable, and privacy-conscious system.

This test plan will also include testing of non-functional requirements including overall system stability, recognition accuracy, privacy protection, data security, and ability to recover from errors due to failed network connections and database issues.

Sprint 1

Sprint 1 focused on defining the system requirements and planning the project. The main features delivered included user stories, the Software Requirements Specification (SRS) with scope, the Work Breakdown Structure (WBS), and the overall project plan.

These features were the focus of Sprint 1 because they establish a clear understanding of what the system will do and how the team will build it. The user stories define system functionality from the user's perspective, while the SRS and scope provide detailed requirements. The WBS and project plan help organize tasks and guide the development process for future sprints.

Sprint 2

Sprint 2 focused on designing and visualizing the system. The main features delivered included the Data Flow Diagram (DFD), ERD diagram, state transition diagram, system architecture diagram, and improved wireframes and prototypes.

These features were the focus of Sprint 2 because they build on the requirements from Sprint 1 and provide a deeper understanding of how the system will function. The diagrams help define system structure, data movement, and behavior, while the prototypes improve the user interface and make the system more interactive and realistic.

Team Members

Resource Name	Role <i>(examples are given below)</i>
Isaiah Bailey	Developer
Moses Ojo	Developer
Gabriel Oniyitan	Developer
Cade Powell	Developer
Austin Woodruff	Developer

1 Scope

The scope of testing for Capstone II includes developing and validating core features. Testing will be grouped by user roles to ensure optimal workflow and features are working correctly.

At the end of Sprint 2, a user must be able to:

1. View system wireframes representing the application layout (US-01)
2. Review the Data Flow Diagram (DFD) showing how data moves through the system (US-02)
3. Review the ERD diagram representing the database structure (US-03)
4. Review the state transition diagram showing system behavior (US-04)
5. Review the system architecture diagram (US-05)
6. Interact with basic prototypes of the application (US-06)

Other testing conducted:

1. Design validation testing

At the end of Sprint 3, a user must be able to:

1. Register an account (US-07)
2. Log in to the system (US-08)
3. View and navigate the instructor dashboard (US-09)

Other testing conducted:

1. Functional testing
2. UI/UX testing

At the end of Sprint 4, a user must be able to:

1. Access the mobile application interface (US-10)
2. Capture their face for enrollment (US-11)
3. Perform attendance check-in using facial recognition (US-12)

Other testing conducted:

1. Integration testing
2. Functional testing
3. Edge Cases
4. Security Testing

At the end of Sprint 5, a user must be able to:

1. Complete the full attendance process from start to finish (US-12)
2. Store and retrieve attendance data (US-13)
3. Handle invalid inputs and system errors (US-14)
4. Use all features without crashes or failures

Other testing conducted:

1. System testing
2. Performance testing
3. Regression testing
4. Edge Cases
5. Privacy testing
6. Usability Testing

7. Assumptions / Risks

Assumptions

1. Users will have consistent internet connectivity during attendance sessions.
2. Mobile devices used for check-in will support camera access and a modern OS.
3. Institutional approval has been granted for Canvas Sandbox integration.
4. Students will provide consent for biometric enrollment where required.
5. The AI recognition model will achieve acceptable accuracy within defined performance thresholds.

Risks

The following risks have been identified, and the appropriate action has been taken to mitigate their impact on the project. The impact (or severity) of the risk is based on how the project would be affected if the risk were triggered. The trigger is what milestone or event would cause the risk to become an issue to be dealt with.

#	Risk	Impact	Trigger	Mitigation Plan
1	Scope Creep – as testers become more familiar with the tool, they will want more functionality	High	New features are requested during sprint 4 and 5	Limit new features to approved requirements. Any other features must be placed in a future improvements list
2	Missed sprint deadlines	Medium	The product did not get delivered on schedule	Track task weekly in trello, any delayed task should be reassigned, reduced in scope, or moved to next sprint
3	Privacy and ethical concerns with stored data	High	Users or instructor question how face data is store	Images from facial recognition will not be stored as raw image files, but instead as embedded files. Also, all data will be stored on the project's local database, not on outside systems like the cloud.
4	Inaccurate facial recognition	Medium	Testing shows facial recognition is inaccurate	A confidence threshold will be implemented to avoid false positives. Users will be instructed to scan their face at multiple angles under good lighting.
5	Performance/processing issues when taking attendance of large classes	Medium	Testing large classes causes crashes or issues with mobile	Ensure there is a large enough upper limit of students whose attendance can be checked per session to ensure even the largest of classes won't overload the application.

			application or the instructor dashboard	
	Firebase connection failure	High	The dashboard and mobile app cannot sync, and attendance records fail to save	Display an error message to prevent a silent failure.

2 Test Approach

The project is using an agile approach, with 2-week sprints. Every sprint will incorporate testing to guarantee ongoing requirements validation and early defect detection. Unit testing, integration testing, and user validation tasks related to finished features will be included in each sprint

Sprint 3

Testing will focus on core system functionality, including user registration, login, and basic dashboard navigation. Unit testing will be used to verify individual components, while functional testing will ensure features work as expected. User validation will confirm that the interface is usable and flows correctly.

Sprint 4

Testing will focus on integrating major features such as the mobile interface and facial recognition for attendance. Integration testing will be used to verify communication between system components. Functional testing will ensure attendance features work correctly, and usability testing will evaluate the user experience.

Sprint 5

Testing will focus on the complete system, including performance, stability, and error stability. System testing and regression testing will be conducted to ensure all features work together without issues. Performance testing will evaluate system reliability under normal usage, and final validation will confirm readiness for deployment.

Test Automation

During each sprint, facial recognition will be verified through manual testing. Manual testing will be used to confirm that face detection and matching are behaving as expected. Integration testing will also be performed manually to verify that recognition is communicating properly with the database and attendance system. As of now, automated testing is not currently being implemented in this version of the project.

Test Cases (Black Box)

2.1.1 Feature 1 (Name this based on the name in your Req doc)

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-001	Successful face scan	FR-05	1. Navigate to camera 2. Ensure face is properly scanned with completion message 3. Check for facial data in local database	Attendance marked present, and attendance data is stored in Firebase, while facial embeddings are stored in the local database
TC-002	Failed face scan	FR-19	1. Navigate to camera 2. Scan face under poor lighting or with a hidden face	Attendance not marked, and face not recognized
TC-003	Late arrival detection	FR-14	1. Navigate to camera 2. Scan face after grace period	Attendance marked but marked late

Test Cases (White Box)

Test Case ID	Description	Test Steps	Expected Output
TC-101	Login	1. Go to login page 2. Enter valid username and password 3. Click "Login"	User is successfully logged in and redirected to dashboard
TC-102	Invalid login	1. Navigate to login page 2. Enter wrong username and password (in separate instances to ensure both work) 3. Click login	Error message showing invalid usernames or password
TC-103	Attendance stored	1. Scan face 2. Ensure user is properly check in on mobile app 3. Check instructor dashboard and Firebase	Instructor can see attendance in real time on dashboard and data is stored in Firebase
TC-104	Manual attendance	1. Sign up for mobile app 2. Opt-out of facial recognition 3. Check in manually	Attendance is manually stored in Firebase and instructors can see manual attendance taken on dashboard
TC-105	Verify students cannot access instructor pages	Test route.page guards with student role	Access is blocked before dashboard data loads
TC-106	Verify late status calculation	Compare check-in time to session start time plus grace period	Status is present before cutoff and late after cutoff

Test Cases

3.3.1..n Name by Feature (follow above table)

Organize the test cases using the table below. Be sure to have a directory in your repository for your test suite that follows a naming scheme that matches the items in the table below.

Like section 3.2, you will add to the table as you progress through each sprint.

Test Case ID	Feature Name	Test Steps	Expected Result	Priority	Notes
3.3.1					
3.3.2					
3.3.3					
...					
3.3.n					

Test Environment

FaceRoll will be tested across multiple environments because it includes multiple services like a mobile app, a facial recognition system, Firebase, and a web-based dashboard for instructors. All of these environments will be unit tested individually and as part of a full system.

The student mobile app will be tested with Expo Go on real mobile devices, which allows the team to verify all features of the app behave properly.

Facial recognition logic, which contains OpenCV, the deepface library, and facenet, which is housed within a Docker container. This container will be tested on multiple computers to

ensure the setup and the logic are correct before moving to the main demo computer for final integration.

Authentication and database testing will be tested with the Firebase emulator suite. This allows testing of login flows and database operations in a controlled environment during development.

The instructor dashboard will be tested in common web browsers using Firebase CLI tools to deploy the website to ensure data loads properly within tables.

Full system testing will combine all systems to verify that the complete workflows function correctly, starting from enrollment to displaying attendance.

Test Schedule

Task Name (<i>sample is below, focus on spring 1 to start</i>)	Start	Finish	Effort	Comments
<i>Test Planning</i>	Sprint 1	Sprint 1	1 week	
<i>Review Requirements documents</i>	Sprint 1	Sprint 1	5 days	
<i>Create initial test estimates</i>	Sprint 1	Sprint 1	2 days	
<i>Learn new test resources</i>	Sprint 1	Sprint 2	1 week	
Setup test environment	Sprint 2	Sprint2	1 week	
<i>Write functional test cases</i>	Sprint 2	Sprint 2	1 week	
<i>Prepare role-based and Ui Test cases</i>	Sprint 2	Sprint 3	4 days	
<i>System testing</i>				
<i>Regression testing</i>				
<i>Usability Testing</i>				
<i>Resolution of final defects and final build testing</i>				
<i>Deploy to Staging environment</i>				
<i>Performance testing</i>				
<i>Release to Production</i>				